

A Appendix: Additional Details for the Indirect-Inference Exercise

This appendix records the details that should not be loaded into the main text. The main section should remain focused on the causal-core message: the model reproduces the output-competition IV response and passes a residual-IV validation. The appendix should instead document the estimator, the optimization path, non-targeted diagnostics, and the reasons why the current run should not yet be interpreted as a full welfare model.

A.1 Estimator and auxiliary moments

The estimation is an indirect-inference exercise because the model is disciplined by auxiliary empirical objects rather than by a closed-form likelihood. For each candidate parameter vector θ , the model generates a simulated panel with the same observation structure as the data. The same auxiliary procedures are then applied to the observed and simulated panels. In the notation of indirect inference, the binding function is

$$h(\theta) = \text{plim } m^s(\theta), \quad (1)$$

where $m^s(\theta)$ is the vector of simulated auxiliary moments. Identification requires that the chosen moments move differently when the economically relevant components of θ are perturbed. In the present run, identification comes from six output-competition moments:

1. the baseline IV response of incumbent markup growth;
2. grouped-IV effects at q80, q85, and q90;
3. the q90 interaction with sectoral concentration; and
4. the sector inverse-markup IV response.

The criterion is

$$Q(\theta) = [\hat{m} - m^s(\theta)]' W [\hat{m} - m^s(\theta)], \quad (2)$$

with W chosen to emphasize statistically stable causal moments. The empirical interpretation of the fitted model therefore depends on the content of these auxiliary moments. It is a model of the output-competition mechanism, not a model of every observed movement in markups.

A.2 Full parameter list

Table 1: Full parameter vector in the causal-core run

Parameter	Value	Status	Interpretation
output_scale	3.449	estimated	Scale of the output-competition channel
output_markup	-0.674	estimated	Baseline output-shock effect on markup growth
high_markup_output	-1.514	estimated	Extra output-shock effect for high-markup firms or sectors
concentration_output	-3.964	estimated	Extra output-shock effect in concentrated markets
share_output	0.500	estimated	Output-shock component in market-share dynamics
share_input	0.000	fixed	Input channel switched off
input_markup	0.000	fixed	Input-markup channel switched off
mean_reversion	1.382	estimated	Pullback of markups toward reference level
drift	-0.104	estimated	Average residual drift
exit_intercept	-2.300	fixed	Baseline exit block fixed
exit_ip	0.000	fixed	No output-shock selection effect
exit_markup	0.000	fixed	No markup-dependent exit effect
eta	1.010	fixed	Elasticity parameter
nu	4.000	fixed	Elasticity parameter
rho	8.000	fixed	Elasticity parameter
lambda_min	0.050	fixed	Lower bound for lambda
lambda_max	0.980	fixed	Upper bound for lambda
default_lambda	0.850	fixed	Default lambda

The fixed zero input and selection parameters are important for interpretation. They explain why the model should be expected to miss input-shock and exit diagnostics. Those misses should guide the next structural extension, not overturn the targeted output-competition result.

A.3 Optimization and targeted fit

The optimizer made 512 objective evaluations. The objective declined from approximately 37.635 to 1.727. This is a substantial improvement and delivers economically interpretable signs: output competition lowers markup growth, and the effect is stronger where market power is initially high. The final objective is not close to zero because the current parameterization cannot perfectly trace the upper-tail gradient and sector aggregation response. This is acceptable for a causal-core run, provided the limitation is made explicit.

Table 2: Moment gaps entering the objective

Target moment	Data	Model	Gap	Weight
baseline_iv_dln_mu	-0.429	-0.441	0.012	50.832
grouped_iv_q80	-0.803	-0.803	-0.001	31.180
grouped_iv_q85	-1.236	-0.942	-0.294	11.768
grouped_iv_q90	-1.625	-1.771	0.146	7.608
grouped_iv_q90_cr4_interaction	-5.371	-5.042	-0.329	0.369
sector_inverse_markup_iv	0.128	0.238	-0.110	40.989

The q85 and q90 pattern is useful diagnostically. The model can generate upper-tail discipline, but the current interaction structure is too rigid to match the exact slope across the upper tail. A future version could allow smoother state dependence in initial markups and concentration rather than relying on a small number of coarse interaction terms.

A.4 Residual-IV validation

The residual-IV validation should be reported near the main SMM results rather than in a generic robustness section. Let

$$r_{ijt} = \Delta \ln \mu_{ijt} - \Delta \ln \mu_{ijt}^{model}. \quad (3)$$

define residual markup growth after removing the fitted output-competition component. Re-estimating the IV regression with r_{ijt} as the dependent variable gives the following validation result.

Table 3: Residual-IV validation

Validation regression	Coef.	SE	Obs.	Clusters	First-stage F
Residual markup growth on output shock	0.042	0.250	31,375	90	22.930

The coefficient is small relative to its standard error and the first stage remains strong. This supports the interpretation that the model absorbs the output-shock variation it was designed to explain. The validation should not be overstated: it does not validate missing input, selection, or decomposition blocks.

A.5 Non-targeted diagnostics

The following diagnostics should be placed in the appendix. They are informative because they show where the model fails once it is asked to explain mechanisms outside the targeted output-competition channel.

Table 4: Selected non-targeted diagnostic moments

Diagnostic moment	Data	Model	Gap	Interpretation
input_shock_between_reallocation	8.055	3.744	4.310	Underpredicts input-shock reallocation
input_shock_within_dln_mu	0.142	-0.306	0.448	Wrong sign for input-markup relation
exit_rate	0.069	0.091	-0.022	Model exit rate too high
exit_iv	0.960	0.324	0.636	Underpredicts output-shock selection response

These misses are not surprising, because the input and selection channels are switched off. They are nevertheless economically important. The data contain an input-shock reallocation pattern and an exit response that the current model cannot reproduce. Therefore, the model should not be used yet for welfare analysis, long-run industry composition, or full manufacturing-wide counterfactuals.

A.6 Counterfactual interpretation

The no-output counterfactual removes the fitted output-competition channel. At the aggregate manufacturing level, the differences are small: the average inverse markup is 0.8361 in the observed model path and 0.8350 in the no-output counterfactual; the average aggregate markup is 1.1966 versus 1.1981. The 2011–2019 change in inverse markups is also close: -0.0183 in the observed path and -0.0192 in the no-output counterfactual.

The right interpretation is therefore local and sectoral. The model says that output competition disciplines exposed incumbent markups, especially in high-rent and concentrated markets. It does not say that Chinese competition explains most aggregate markup dynamics in Turkish manufacturing. This distinction is a strength of the exercise because it prevents the model from being used beyond the variation that identifies it.

A.7 Recommended reporting allocation

The main text should report only the objects needed for the causal-core claim:

1. the motivation for using a model after IV;
2. the SMM/indirect-inference criterion;
3. the compact parameter table;
4. the targeted causal moment fit;
5. the residual-IV validation; and
6. a restrained counterfactual synthesis.

The appendix should report the full parameter vector, optimization history, full moment gaps, non-targeted diagnostics, input and selection failures, sector-level counterfactual heterogeneity, and allocative-wedge checks. The final thesis claim should remain: the exercise supports an output-competition markup-discipline channel, not a complete welfare model of Turkish manufacturing.